MACHINE LEARNING-BASED CLASSIFICATION OF BREAST TUMOR AND ADJACENT NON-TUMOR TISSUE USING RNA-seq DATA

A reproducible transcriptomic classification and feature-interpretation study

Course Project: Data Science and Cloud Computing in Bioinformatics

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Programme: MSc Bioinformatics

Institution: Addis Ababa University

Dataset: NCBI GEO GSE233242

Computing: Python, scikit-learn, XGBoost, Snakemake, SLURM/HPC

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Abstract

This project evaluated whether RNA-seq gene-expression profiles could distinguish breast tumor from adjacent non-tumor tissue and identified model-associated genes for exploratory biological interpretation. The NCBI GEO series GSE233242 supplied a gene-count matrix with 39,376 genes and 84 samples. Metadata reconciliation yielded 42 tumor and 42 normal samples. Filtering for CPM ≥1 in at least 17 samples retained 18,863 genes, which were transformed as log2(CPM+1). Random Forest and XGBoost were compared using patient-grouped five-fold StratifiedGroupKFold cross-validation, with feature selection conducted inside each fold. Random Forest with 500 genes was selected by mean ROC-AUC (0.967 ± 0.047). Held-out test metrics were all 1.00, but the test set comprised only 16 samples from eight patient pairs, so this result is preliminary and requires external validation. Feature selection identified 248 genes in all five folds and 321 in at least four folds. Nineteen consensus genes showed significant paired tumor-associated expression differences across 41 complete patient pairs after Benjamini–Hochberg correction. Exploratory enrichment highlighted actin/cytoskeleton terms among tumor-down genes, while tumor-up terms were heterogeneous. The workflow demonstrates reproducible computational analysis, but the model-derived genes are not validated biomarkers.

Keywords: breast cancer; RNA-seq; machine learning; Random Forest; feature selection; paired validation; pathway enrichment; Snakemake.

1. Introduction

Breast cancer is molecularly heterogeneous, and RNA sequencing measures gene-expression patterns across the transcriptome. Machine-learning methods can exploit multivariate expression signals for tumor-versus-normal classification, but high-dimensional data require careful control of feature-selection bias, sample leakage, and overinterpretation of model importance.

This study combines patient-level splitting, fold-contained feature selection, comparative model evaluation, feature-stability analysis, paired expression testing, and exploratory enrichment. The goal is to demonstrate an interpretable and reproducible classification workflow, not to establish a clinical diagnostic assay.

2. Research Question and Objectives

Research question: Can gene-expression profiles distinguish breast tumor from adjacent non-tumor tissue, and which genes contribute consistently to that classification?

2.1 General objective

Develop and evaluate a reproducible machine-learning workflow for classifying breast tumor and adjacent non-tumor tissue from RNA-seq gene-expression data.

2.2 Specific objectives

- Reconcile GEO count data and sample metadata.
- Filter low-expression genes, transform expression values, and explore sample variation.
- Compare Random Forest and XGBoost using patient-grouped cross-validation.
- Evaluate the selected model on a patient-disjoint test set.
- Quantify feature-selection stability and construct a consensus set.
- Assess paired tumor-normal expression differences and explore functional enrichment.
- Automate analysis dependencies with Snakemake and execute on HPC/SLURM.

3. Materials and Methods

3.1 Dataset and metadata

The analysis used NCBI GEO series GSE233242 and the files GSE233242_raw_counts_GRCh38.p13_NCBI.tsv.gz and GSE233242_series_matrix.txt.gz. The matrix contained 39,376 genes and 84 samples, with 42 tumor and 42 normal samples. Tumor annotations included 29 Luminal A, nine TNBC, three Luminal B, and one HER2 sample; the analysis was therefore not restricted to Luminal A. Two metadata records lacked corresponding count-matrix columns: GSM7416151 (Tumor 19) and GSM7416157 (Normal 22). For paired analyses, 41 complete patient pairs (82 samples) were available. The two singleton samples were excluded from paired analyses only.

3.2 Filtering, transformation, and EDA

Raw library sizes ranged from approximately 2.46 million to 106.7 million counts, with a median of approximately 25.53 million. Genes were retained when CPM ≥1 in at least 17 samples, retaining 18,863 genes. Expression was transformed as log2(CPM+1). PCA showed PC1 explaining 31.07% and PC2 explaining 13.52% of variance; the first two components explained 44.58%, and the first ten approximately 74%. PCA indicated broad normal/tumor separation and tumor heterogeneity. No outliers were removed.

3.3 Patient-grouped split and model evaluation

GroupShuffleSplit with random_state=42 created a training partition of 33 patient pairs (66 samples) and a test partition of eight patient pairs (16 samples). No patient appeared in both partitions. Random Forest and XGBoost were compared using five-fold StratifiedGroupKFold. SelectKBest with the ANOVA F-statistic was fitted within each training fold for candidate feature counts k=100, 250, 500, and 1,000. Both classifiers used 300 trees/estimators. Selection prioritized mean cross-validation ROC-AUC, then F1. Metrics were accuracy, precision, recall, F1, and ROC-AUC.

3.4 Feature stability and paired validation

Feature selection was recorded across the five folds. A consensus set combined high Random Forest feature importance, repeated selection, and the final selected feature set. Nineteen consensus genes were assessed using Wilcoxon signed-rank tests on 41 complete patient pairs, with Benjamini–Hochberg FDR correction. Differences are on the log2(CPM+1) scale; 2 raised to a median difference is not a formal count-based fold change.

3.5 Enrichment and workflow

Tumor-up and tumor-down consensus genes were analyzed separately through the g:Profiler REST API, using human ENTREZ Gene identifiers, GO:BP/MF/CC, Reactome and KEGG sources, and the 18,863 filtered genes as background. Given the small query lists, enrichment was interpreted as exploratory. Snakemake 7.32.4 orchestrated modeling, feature stability, annotation, paired validation, plots, and enrichment. Execution used SLURM on betinserver with two CPUs and 4 GB memory, using Python 3.10.22 in the project2_ml environment.



Figure 1. Workflow overview based on the audited project stages.

4. Results

4.1 Dataset summary

Characteristic	Result
Samples	84: 42 tumor, 42 normal
Genes before filtering	39,376
Genes retained	18,863
Complete pairs	41 (82 samples)
Training partition	33 patient pairs; 66 samples
Test partition	8 patient pairs; 16 samples
Median library size	~25.53 million counts
PCA PC1 / PC2 / PC1+PC2	31.07% / 13.52% / 44.58%

4.2 Cross-validation

Random Forest with 500 genes had the highest mean ROC-AUC and was selected. XGBoost with 250 genes had somewhat higher mean accuracy and F1, but a lower mean ROC-AUC.

Model	Genes	Accuracy	Precision	Recall	F1	ROC-AUC ± SD
Random Forest	500	0.867	0.900	0.848	0.864	0.967 ± 0.047
XGBoost	250	0.883	0.900	0.881	0.886	0.963 ± 0.062
Random Forest	250	0.867	0.900	0.848	0.864	0.963 ± 0.051
XGBoost	100	0.883	0.892	0.881	0.882	0.961 ± 0.066
Random Forest	1000	0.867	0.900	0.848	0.864	0.956 ± 0.051

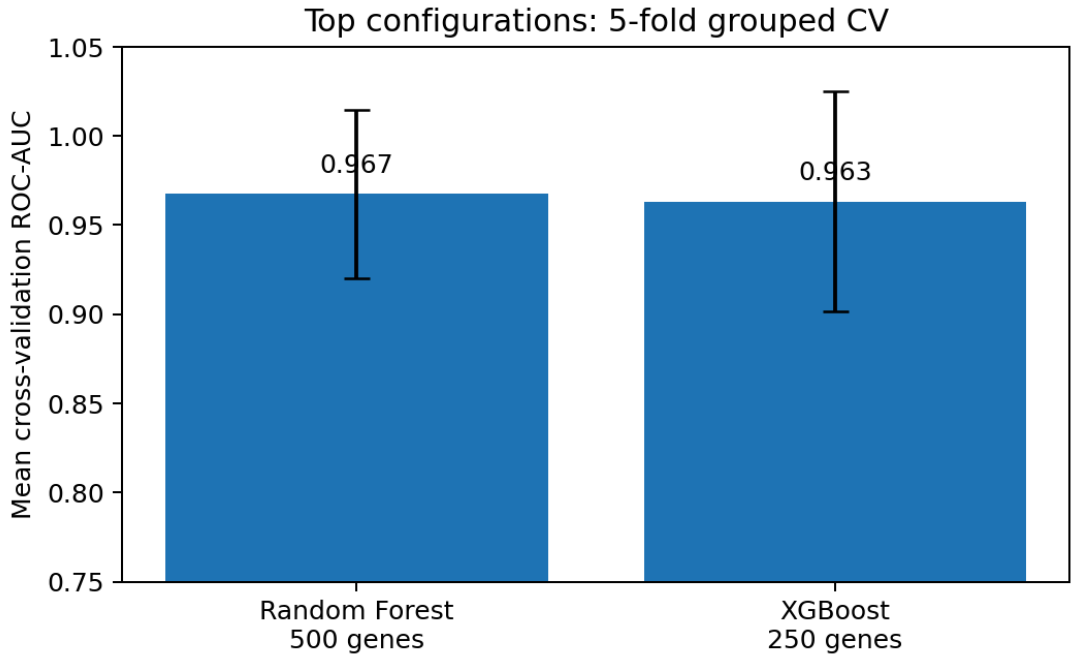


Figure 2. Mean ROC-AUC for the leading configurations; error bars show reported fold SD.

4.3 Held-out test set

Metric	Result
Accuracy	1.000
Precision	1.000
Recall	1.000
F1	1.000
ROC-AUC	1.000
Test size	16 samples from 8 patient pairs

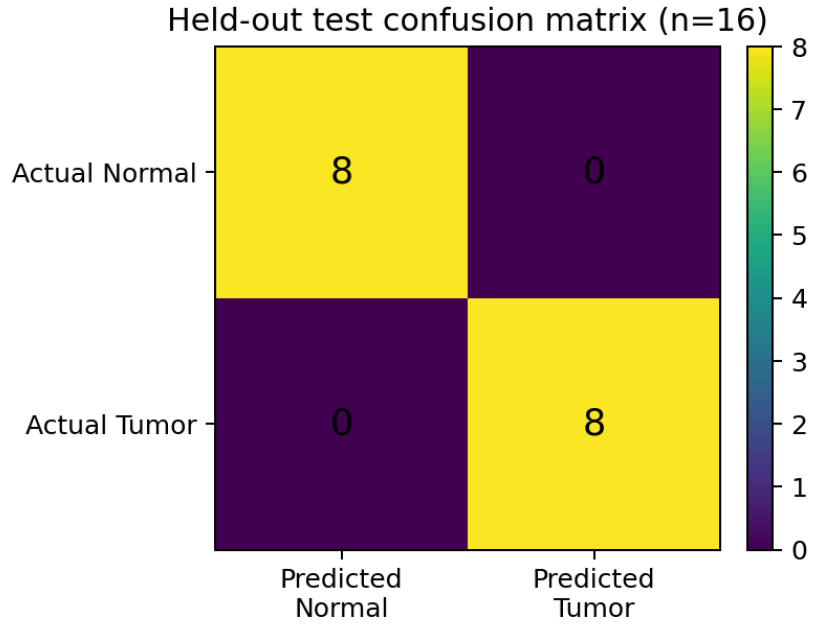


Figure 3. Confusion matrix implied by the audited test results: eight normal and eight tumor samples correctly classified.

The perfect test metrics are preliminary. With only 16 test samples, one error would change accuracy by 6.25 percentage points. External validation on a larger independent cohort is essential.

4.4 Feature stability and consensus genes

A total of 248 genes were selected in all five folds and 321 in at least four folds. The final consensus set contained 19 genes. All 19 showed statistically significant paired tumor-associated expression differences after FDR correction across 41 complete patient pairs.

Gene	Direction	Median paired difference	FDR
KIF26B	Up	+3.259	7.20×10^{-11}
PYCR1	Up	+2.385	1.52×10^{-10}
LEF1	Up	+1.797	8.15×10^{-11}
MBOAT2	Up	+1.635	1.51×10^{-10}
MGAT4A	Up	+1.297	1.52×10^{-10}
PACC1	Up	+1.041	3.12×10^{-10}
PPP1R14A	Down	−2.824	5.45×10^{-10}
GSN	Down	−2.317	1.97×10^{-10}
CA4	Down	−2.247	7.20×10^{-11}
KIF26A	Down	−2.201	3.12×10^{-10}
GeneID_100532742	Down	−2.191	2.59×10^{-11}
KANK3	Down	−2.109	7.20×10^{-11}
LMOD1	Down	−1.869	1.04×10^{-8}
ADCYAP1R1	Down	−1.684	3.54×10^{-10}
CYGB	Down	−1.475	1.73×10^{-10}
PNMA8B	Down	−1.301	7.20×10^{-11}
WTIP	Down	−1.136	6.34×10^{-9}
NKAPL	Down	−0.943	2.59×10^{-11}
SNX33	Down	−0.624	5.45×10^{-10}

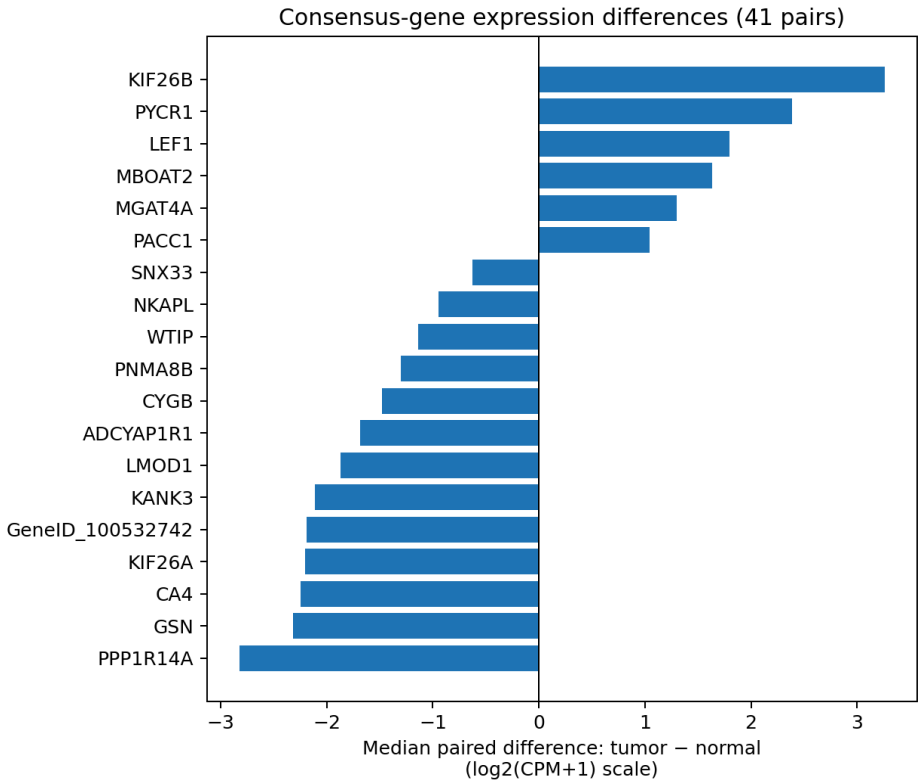


Figure 4. Median paired tumor-minus-normal differences on the log2(CPM+1) scale, calculated from audited summary statistics.

KIF26B had the largest positive median paired difference (+3.259), followed by PYCR1 (+2.385). PPP1R14A showed the largest negative difference (−2.824). These are model-associated features with consistent tumor-associated differences, not validated biomarkers. Paired validation was post hoc and included test-partition pairs, so it is not independent validation of predictive performance.

4.5 Functional enrichment

Tumor-down enrichment was relatively coherent around actin filament polymerization, regulation of actin polymerization/depolymerization, actin filament length, cytoskeleton organization, and protein polymerization. Several leading terms intersected three of the twelve tumor-down query genes. Tumor-up enrichment was more heterogeneous, including alanine-glyoxylate transaminase activity, pH-gated chloride channel activity, pyrroline-5-carboxylate reductase activity, glycoprotein-processing enzyme activity, and acylglycerophospholipid O-acyltransferase-related terms. Many tumor-up terms were supported by only one of six query genes. These are exploratory enrichment terms, not proof of broad pathway alteration.

5. Discussion

5.1 Model performance

The mean cross-validation ROC-AUC of 0.967 indicates strong discrimination in this cohort. The grouped split and fold-contained feature selection are methodological strengths because they reduce patient leakage and feature-selection bias. The held-out score of 1.00 should nevertheless be interpreted cautiously because the test set is very small and a single partition cannot fully characterize uncertainty.

5.2 Biological interpretation

Repeated feature selection supports a degree of stability within this dataset. Paired testing further showed that the consensus genes differed consistently between tumor and normal tissue. However, the paired analysis was post hoc and included samples from the test partition; it supports expression association within this cohort, not independent validation of the classifier. The actin/cytoskeleton enrichment among tumor-down genes is hypothesis-generating. Tumor-up enrichment is less coherent and dominated by terms with small intersections.

5.3 Strengths and limitations

- Strengths: patient-level split; feature selection inside CV; feature stability analysis; paired tests with FDR correction; Snakemake/HPC workflow.
- Small held-out test set (16 samples) and no external validation.
- Multiple tumor subtypes were included; target was tumor versus normal, not subtype classification.
- Post-hoc paired expression validation used the full paired cohort, including test pairs.
- Consensus genes are model-derived features, not validated clinical biomarkers.
- Small enrichment query sets can yield unstable terms.
- log2(CPM+1) supports this modeling workflow but is not a count-based differential-expression model.

6. Reproducibility and outputs

The audited Snakemake dry run returned “Nothing to be done (all requested files are present and up to date).” Representative outputs include:

Output group	Representative outputs
EDA	results/eda/pca_tumor_normal.png; pca_subtype.png; sample_correlation_heatmap.png; library_sizes.tsv
ML metrics	results/ml/cv_summary.tsv; cv_fold_results.tsv; test_metrics.tsv; test_predictions.tsv; roc_curve.png; confusion_matrix.png
Features	final_selected_genes.tsv; feature_importance.tsv; cv_feature_selection.tsv; feature_selection_stability.tsv; stable_genes.tsv; consensus_genes.tsv
Paired validation	consensus_gene_annotation.tsv; consensus_gene_expression_validation.tsv; final_consensus_gene_summary.tsv; consensus_gene_heatmap.png; paired_top_consensus_genes.png
Enrichment	results/pathways/tumor_up_enrichment.tsv; tumor_down_enrichment.tsv; pathway_enrichment_summary.tsv; pathway_report_table.tsv
Models/workflow	models/final_feature_selector.joblib; models/final_model.joblib; hpc/run_project2.sbatch; Snakemake configuration and logs

Repository: https://github.com/dejeneseboka094-coder/Data-Science-and-Cloud-Computing-in-Bioinformatics. Before submission, confirm the repository contains the final scripts, configuration, and documentation.

7. Conclusion

This project developed a patient-aware, reproducible machine-learning workflow for classifying breast tumor and adjacent non-tumor tissue from public RNA-seq count data. Random Forest with 500 selected genes was chosen by mean cross-validation ROC-AUC (0.967 ± 0.047). Perfect held-out test metrics were observed in a small test set and require independent validation. Feature selection showed stability, and 19 consensus genes exhibited significant paired tumor-associated expression differences across 41 complete pairs. Tumor-down enrichment suggested actin/cytoskeleton-related functions, whereas tumor-up enrichment was heterogeneous. The project demonstrates an end-to-end computational workflow while avoiding unsupported biomarker or clinical-readiness claims.

8. Recommendations

- Validate a locked model in a larger independent cohort with compatible preprocessing.
- Use repeated patient-level resampling and uncertainty intervals to better quantify performance.
- Evaluate calibration, class-specific errors, and subtype-specific performance.
- Validate consensus genes and enrichment hypotheses independently.
- Use a count-aware statistical framework if formal differential-expression analysis becomes a project objective.
- Archive package versions and ensure scripts, configuration, and instructions are version-controlled.

References

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Appendix: Figure provenance and interpretation notes

Figures in this report were generated from the audited numeric summaries and workflow description. The exact original PCA plots, expression heatmap, paired sample plots, and ROC curve are also present in the project output inventory: results/eda/pca_tumor_normal.png, results/eda/pca_subtype.png, results/ml/consensus_gene_heatmap.png, results/ml/paired_top_consensus_genes.png, and results/ml/roc_curve.png. The original image binaries were not attached with the audit text, so they were not recreated from unavailable sample-level data.

The reported 2^(median paired difference) values are approximate representations from the log2(CPM+1) scale and should not be labeled formal count-based fold changes.